

Question ID: 55cfaf22

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Data set X: **5, 9, 9, 13**

Data set Y: **5, 9, 9, 13, 27**

The lists give the values in data sets X and Y. Which statement correctly compares the mean of data set X and the mean of data set Y?

Answer

- A. The mean of data set X is greater than the mean of data set Y.
- B. The mean of data set X is less than the mean of data set Y.
- C. The means of data set X and data set Y are equal.
- D. There is not enough information to compare the means.

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values in the data set divided by the number of values in the data set. It follows that the mean of data set X is $\frac{5+9+9+13}{4}$, or **9**, and the mean of data set Y is $\frac{5+9+9+13+27}{5}$, or **12.6**. Since **9** is less than **12.6**, the mean of data set X is less than the mean of data set Y.

Alternate approach: Data set Y consists of the **4** values in data set X and one additional value, **27**. Since the additional value, **27**, is larger than any value in data set X, the mean of data set X is less than the mean of data set Y.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.
- Choice D is incorrect and may result from conceptual or calculation errors.